Sahaj Bhadja

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Summary

Python-focused software engineer pursuing an MSCS with 2+ years building **backend tooling**, data workflows, CI/CD pipelines, Linux-ready services, automated tests, and reliable internal platforms.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Operating Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **Python backend tooling** for 500+ users, translating requirements into APIs, normalized schemas, monitored jobs, validation logic, documentation, CI/CD-backed production releases, support handoffs, Linux readiness, and reviews.
- Improved **data platform performance** by tuning indexed queries, background jobs, and deployment workflows, reducing dashboard latency from **1.2s to 350ms** while supporting reliable analytics, monitoring, and reviews.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cross-platform Python and SQL services** using AWS Glue and S3, improving response time by **30%** across integrations, release reviews, operational handoffs, data workflows, fixes, and deployments.
- Maintained **20+ automated test suites** with Jenkins pipelines, PyTest, and Mockito across validation flows, increasing CI reliability by **68%** while reducing release regressions, defects, rework, and alerts.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Python services and REST APIs** handling 10,000+ daily requests, implementing file ingestion, concurrent processing, automated checks, documented fixes, sprint delivery, triage, release reviews, and support.
- Partnered with users to debug **production workflow defects** by reproducing OCR, persistence, and execution failures, documenting regression cases, reviewing fixes, and cutting feedback turnaround by **60%** safely.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **modular internal services** with REST APIs, WebSockets, and SQL/NoSQL persistence, supporting 2M+ synchronized records across dashboards, integrations, monitoring, debugging, validation, reports, documentation, tooling, releases, and reviews.
- Automated **Docker-based build and test flows** with Jenkins, cutting release time by **50%** while improving deployment consistency, troubleshooting documentation, shared-environment handoffs, pipeline reliability, checks, releases, and reports.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python evaluation pipelines** across 100,000+ records, producing structured metrics, logs, and visual summaries for reproducible experiments, model-analysis workflows, diagnostics, reporting, analysis reviews, dashboards, and summaries.
- Documented **prompt, evaluation, and reporting workflows** for researchers, turning ad hoc experiments into repeatable processes with clearer technical handoffs, validation outputs, analysis notes, summaries, reviews, and reports.

Skills

Languages: Python 3, JavaScript, TypeScript, Java, SQL, C++, Bash, HTML/CSS
Backend: Django familiarity, Flask/FastAPI, REST APIs, service design, validation, internal tooling, bug fixes
Data: PostgreSQL, MySQL, SQL Server, MongoDB, ETL, indexing, query tuning, logs, data pipelines
DevOps: Linux, Jenkins pipelines, Docker, Git, GitHub, CI/CD, Jira-style Agile, deployment support
Quality: PyTest, Mockito, code reviews, documentation, troubleshooting, production support, secure data workflows

Projects

Linguistix Code | *Python, AWS, Docker, CI/CD*

- Developed a **containerized Python REST API** with AWS deployment, CI/CD releases, validation checks, and reproducible execution, achieving **87%** classification accuracy with diagnostic reporting, automated tests, and logs.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built a **document retrieval platform** over 50K+ documents using transformer embeddings and FAISS, enabling contextual Q&A, summarization, access controls, optimized retrieval, user documentation, and search workflows.

Chat2SQL Code | *Java, TypeScript, SQL, Docker*

- Built an **AI-powered SQL workspace** supporting 10+ databases, accelerating query authoring, schema exploration, and database management through Java services, TypeScript interfaces, Docker workflows, validation, reviews, and tests.